Lathi 20题练习册 第2批

对应 Boulet 前四章

原书英文题干、公式与必要图示 · 无答案版

记号：D = d/dt；差分式中 E 为前移算子，E{y[n]} = y[n+1]；j^2 = -1
u 为单位阶跃，δ 为单位冲激；LTIC / LTID 分别指连续 / 离散时间 LTI 系统

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来源：B. P. Lathi / R. A. Green, Linear Systems and Signals, 3rd ed. (2018)
中文主题仅用于课程对应；题目要求以原书英文为准。

能量与幅度、时间平移及尺度变换

Lathi 1.1-9 | 原书第 137、138 页

Boulet 第1章：Finite-Energy and Finite-Power Signals，p.21；Transformations of the Time Variable，p.4

1.1-9 There are many useful properties related to signal energy. Prove each of the following statements. In each case, let energy signal $x_1(t)$ have energy $E[x_1(t)]$, let energy signal $x_2(t)$ have energy $E[x_2(t)]$, and let T be a nonzero, finite, real-valued constant.

(a) Prove $E[Tx_1(t)] = T^2E[x_1(t)]$. That is, amplitude scaling a signal by constant T scales the signal energy by T^2.

(b) Prove $E[x_1(t)] = E[x_1(t - T)]$. That is, shifting a signal does not affect its energy.

(c) If $(x_1(t) \neq 0) \Rightarrow (x_2(t) = 0)$ and $(x_2(t) \neq 0) \Rightarrow (x_1(t) = 0)$, then prove $E[x_1(t) + x_2(t)] = E[x_1(t)] + E[x_2(t)]$. That is, the energy of the sum of two nonoverlapping signals is the sum of the two individual energies.

(d) Prove $E[x_1(Tt)] = (1/|T|)E[x_1(t)]$. That is, time-scaling a signal by T reciprocally scales the signal energy by $1/|T|$.

演算区

阶跃与冲激的积分、有限能量和信号支集

Lathi 1.4-4 | 原书第 141 页

Boulet 第1章：Generalized Functions，p.26；Finite-Energy and Finite-Power Signals，p.21

1.4-4 Define signal $x(t) = u(t-1) - u(t-2.5) - 2\delta(t-4) + \delta(t-6)$.

(a) Sketch $y(t) = \int_{-\infty}^{t} x(\tau)\,d\tau$.

(b) Describe a simple change that can be made to the right-most delta function in $x(t)$ so that $y(t) = \int_{-\infty}^{t} x(\tau)\,d\tau$ has finite energy.

(c) Sketch $z(t) = \int_{t}^{\infty} x(\tau)\,d\tau$.

(d) Determine real constants A and B so that $w(t) = x\left(\frac{t-A}{B}\right)$ has a region of support $[-2, 2]$.

演算区

连续信号的奇偶分解与能量分配

Lathi 1.5-3 | 原书第 143 页

Boulet 第1章：Even and Odd Signals, p.23；Finite-Energy and Finite-Power Signals, p.21

1.5-3 (a) Determine even and odd components of the signal $x(t) = e^{-2t}u(t)$.

(b) Show that the energy of $x(t)$ is the sum of energies of its odd and even components found in part (a).

(c) Generalize the result in part (b) for any finite energy signal.

演算区

时间不变系统与时变系统的判断

Lathi 1.7-2 | 原书第 144 页

Boulet 第1章：System Models and Basic Properties, p.34

1.7-2 For the systems described by the following equations, with the input $x(t)$ and output $y(t)$, explain with reasons which of the systems are time-invariant parameter systems and which are time-varying-parameter systems.

(a) $y(t) = x(t-2)$

(b) $y(t) = x(-t)$

(c) $y(t) = x(at)$

(d) $y(t) = t\,x(t-2)$

(e) $y(t) = \int_{-5}^{5} x(\tau)\,d\tau$

(f) $y(t) = \left(\frac{dx(t)}{dt}\right)^2$

演算区

离散信号的奇偶分解

Lathi 3.3-9 | 原书第 317 页

Boulet 第1章：Even and Odd Signals，p.23；Discrete-Time Impulse and Step Signals，p.25

题干提到第1.5节，是说明奇偶分解概念与连续时间相同；不依赖其他习题。

3.3-9 The concepts of even and odd functions for discrete-time signals are identical to those of the continuous-time signals discussed in Sec. 1.5. Using these concepts, find and sketch the odd and the even components of the following:

(a) $u[n]$

(b) $nu[n]$

(c) $\sin\left(\frac{\pi n}{4}\right)$

(d) $\cos\left(\frac{\pi n}{4}\right)$

演算区

卷积结果的奇偶性

Lathi 2.4-11 | 原书第 227 页

Boulet 第2章：Continuous-Time LTI Systems: The Convolution Integral，p.67；Properties of Linear Time-Invariant Systems，p.74

提示所引用2.4-10的性质：若 x(t)*g(t)=c(t)，则 x(at)*g(at)=c(at)/|a|（a != 0）。

2.4-11 Show that the convolution of an odd and an even function is an odd function and the convolution of two odd or two even functions is an even function. [*Hint:* Use the time-scaling property of convolution in Prob. 2.4-10.]

演算区

阶跃、重复指数与斜坡的连续卷积

Lathi 2.4-14 | 原书第 227 页

Boulet 第2章：Continuous-Time LTI Systems: The Convolution Integral, p.67

2.4-14 Using direct integration, find $u(t) * u(t)$, $e^{-at}u(t) * e^{-at}u(t)$, and $tu(t) * u(t)$.

演算区

含冲激的连续系统零状态响应

Lathi 2.4-21 | 原书第 227 页

Boulet 第2章：Continuous-Time LTI Systems: The Convolution Integral，p.67

2.4-21 A first-order allpass filter impulse response is given by

$$h(t) = -\delta(t) + 2e^{-t}u(t)$$

(a) Find the zero-state response of this filter for the input $e^{t}u(-t)$.
(b) Sketch the input and the corresponding zero-state response.

演算区

含时移指数和冲激的离散卷积

Lathi 3.8-6 | 原书第 322 页

Boulet 第2章：Discrete-Time LTI Systems: The Convolution Sum， p.54

3.8-6 Find the (zero-state) response $y[n]$ of an LTID system if the input is $x[n] = 3^{n-1}u[n+2]$, and

$$h[n] = \tfrac{1}{2}[\delta[n-2] - (-2)^{n+1}]u[n-3]$$

演算区

离散指数输入与卷积时移

Lathi 3.8-16 | 原书第 322 页

Boulet 第2章：Discrete-Time LTI Systems: The Convolution Sum，p.54；Properties of Linear Time-Invariant Systems，p.74

提示中的式(3.32)为卷积的时移性质：若 x[n]*h[n]=y[n]，则 x[n-n0]*h[n]=y[n-n0]。

3.8-16 Find an LTID system (zero-state) response if its impulse response $h[n] = (0.5)^n u[n]$, and the input $x[n]$ is

(a) $2^n u[n]$

(b) $2^{n-3} u[n]$

(c) $2^n u[n-2]$

[*Hint:* You may need to use the convolution shift property of Eq. (3.32).]

演算区

微分系统的特征模态与零输入响应

Lathi 2.2-3 | 原书第 224 页

Boulet 第3章：Causal LTI Systems Described by Differential Equations，p.92；Characteristic Polynomials and Stability，p.112

2.2-3 A real LTIC system with input $x(t)$ and output $y(t)$ is described by the following constant-coefficient linear differential equation:

$$(D^3 + 9D)\{y(t)\} = (2D^3 + 1)\{x(t)\}.$$

(a) What is the characteristic equation of this system?
(b) What are the characteristic modes of this system?
(c) Assuming $y_{zir}(0) = 4$, $\dot{y}_{zir}(0) = -18$, and $\ddot{y}_{zir}(0) = 0$, determine this system's zero-input response $y_{zir}(t)$. Simplify $y_{zir}(t)$ to include only real terms (i.e., no j's should appear in your answer).

演算区

二阶微分系统与初始条件响应

Lathi 2.2-4 | 原书第 224 页

Boulet 第3章：Causal LTI Systems Described by Differential Equations，p.92

2.2-4 An LTIC system is specified by the equation

$$(D^2+5D+6)y(t)=(D+1)x(t)$$

(a) Find the characteristic polynomial, characteristic equation, characteristic roots, and characteristic modes of this system.
(b) Find $y_0(t)$, the zero-input component of the response $y(t)$ for $t \geq 0$, if the initial conditions are $y_0(0^-)=2$ and $\dot{y}_0(0^-)=-1$.

演算区

差分方程的前三项递推求解

Lathi 3.5-2 | 原书第 319 页

Boulet 第3章：Causal LTI Systems Described by Difference Equations， p.96

3.5-2 Solve recursively (first three terms only):

(a) $y[n+1]-0.5y[n]=0$, with $y[-1]=10$

(b) $y[n+1]+2y[n]=x[n+1]$, with $x[n]=e^{-n}u[n]$ and $y[-1]=0$

演算区

非零初始条件下的离散系统全响应

Lathi 3.8-15 | 原书第 322 页

Boulet 第3章：Causal LTI Systems Described by Difference Equations，p.96

与本批第13题的(b)小题使用相同方程及输入，但本题初值为 y[-1]=10，第13题(b)为 y[-1]=0；本题要求全响应，而第13题只要求递推前三项。

3.8-15 Find the total response of a system specified by the equation

$$y[n+1]+2y[n]=x[n+1]$$

if $y[-1]=10$, and the input $x[n]=e^{-n}u[n]$.

演算区

离散系统的BIBO稳定性与内部稳定性

Lathi 3.9-2 | 原书第 327 页

Boulet 第3章：Characteristic Polynomials and Stability of Differential and Difference Systems, p.112

3.9-2 Each of the following equations specifies an LTID system. Determine whether each of these systems is BIBO-stable or -unstable. Determine also whether each is asymptotically stable, unstable, or marginally stable.

(a) $y[n+2]+0.6y[n+1]-0.16y[n]=x[n+1]-2x[n]$

(b) $y[n]+3y[n-1]+2y[n-2]=x[n-1]+2x[n-2]$

(c) $(E-1)^2\left(E+\frac{1}{2}\right)y[n]=x[n]$

(d) $y[n]+2y[n-1]+0.96y[n-2]=x[n]$

(e) $y[n]+y[n-1]-2y[n-2]=x[n]+2x[n-1]$

(f) $(E^2-1)(E^2+1)y[n]=x[n]$

演算区

三角与复指数傅里叶频谱的相互转换

Lathi 6.3-3 | 原书第 672 页

Boulet 第4章：Determination of the Fourier Series Representation，p.134；Graph of the Fourier Series Coefficients: The Line Spectrum，p.137

6.3-3 A periodic signal $x(t)$ is expressed by the following Fourier series:

$$x(t) = 3\cos t + \sin\left(t - \frac{\pi}{6}\right) - 2\cos\left(t - \frac{\pi}{3}\right)$$

(a) Sketch the amplitude and phase spectra for the trigonometric series.
(b) By inspection of spectra in part (a), sketch the exponential Fourier series spectra.
(c) By inspection of spectra in part (b), write the exponential Fourier series for $x(t)$.
(d) Show that the series found in part (c) is equivalent to the trigonometric series for $x(t)$.

演算区

由复指数级数求三角频谱与带宽

Lathi 6.3-5 | 原书第 672、673 页

Boulet 第4章：Graph of the Fourier Series Coefficients: The Line Spectrum， p.137

6.3-5 The exponential Fourier series of a certain function is given as

$$x(t) = (2+j2)e^{-j3t} + j2e^{-jt} + 3 - j2e^{jt} + (2-j2)e^{j3t}$$

(a) Sketch the exponential Fourier spectra.
(b) By inspection of the spectra in part (a), sketch the trigonometric Fourier spectra for $x(t)$. Find the compact trigonometric Fourier series from these spectra.
(c) Show that the trigonometric series found in part (b) is equivalent to the exponential series for $x(t)$.
(d) Find the signal bandwidth.

演算区

周期信号时移和时间尺度变换的频谱性质

Lathi 6.3-10 | 原书第 674 页

Boulet 第4章：Properties of Continuous-Time Fourier Series, p.139

6.3-10 A periodic signal $x(t)$ is expressed as an exponential Fourier series

$$x(t) = \sum_{n=-\infty}^{\infty} D_n e^{jn\omega_0 t}$$

(a) Show that the exponential Fourier series for $\hat{x}(t) = x(t-T)$ is given by

$$\hat{x}(t) = \sum_{n=-\infty}^{\infty} \hat{D}_n e^{jn\omega_0 t}$$

in which

$$|\hat{D}_n| = |D_n| \quad \text{and} \quad \angle \hat{D}_n = \angle D_n - n\omega_0 T$$

This result shows that time shifting of a periodic signal by T seconds merely changes the phase spectrum by $n\omega_0 T$. The amplitude spectrum is unchanged.

(b) Show that the exponential Fourier series for $\tilde{x}(t) = x(at)$ is given by

$$\tilde{x}(t) = \sum_{n=-\infty}^{\infty} D_n e^{jn(a\omega_0)t}$$

This result shows that time compression of a periodic signal by a factor a expands its Fourier spectra along the ω axis by the same factor a. Similarly, time expansion of a periodic signal by a factor a compresses its Fourier spectra along the ω axis by the factor a. Intuitively explain this result.

谐波分量通过低通、高通、带通与带阻滤波器

Lathi 6.4-2 | 原书第 675 页

Boulet 第4章：Steady-State Response of an LTI System to a Periodic Signal, p.155

6.4-2 A periodic signal $x(t) = 1 + 2\cos(5\pi t) + 3\sin(14\pi t)$ is applied to an LTIC system to produce output $y(t)$.

(a) Determine ω_0, the fundamental radian frequency of $x(t)$.

(b) Determine D_n, the exponential Fourier series spectrum of $x(t)$.

(c) If the system is an ideal lowpass filter with cutoff frequency $f_c = 2$ Hz, what is the output $y(t)$?

(d) If the system is an ideal highpass filter with cutoff frequency $f_c = 2$ Hz, what is the output $y(t)$?

(e) If the system is an ideal bandpass filter with a 4 Hz passband centered at 4 Hz, what is the output $y(t)$?

(f) If the system is an ideal bandstop filter with a 5 Hz stopband centered at 10 Hz, what is the output $y(t)$?

(g) Describe the frequency response of a filter that, in response to $x(t)$, would produce the output $y(t) = 4\cos(5\pi t) - 9\sin(14\pi t)$.

演算区

同一有限区间信号的不同傅里叶级数表示

Lathi 6.5-7 | 原书第 677 页

Boulet 第4章：Determination of the Fourier Series Representation，p.134；Optimality and Convergence of the Fourier Series，p.144

6.5-7 Represent $x(t) = t$ over the interval $(0, 1)$ by a trigonometric Fourier series that has

(a) $\omega_0 = 2\pi$ and only sine terms

(b) $\omega_0 = \pi$ and only sine terms

(c) $\omega_0 = \pi$ and only cosine terms

You may use a dc term in these series if necessary.

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